W2 PRACTICE

MOBILE ***UI HEURISTICS***

In this research work, you will work on a **specific mobile interaction scenario** in terms of UI heuristics

- ✓ This work needs to be **performed and presented in pairs**.

Learning objectives

- ✓ Develop a **deep understanding of Mobile UI heuristics**
- ✓ Take the **right decisions** in terms of **UI components** - given a real-world problem

How to submit?

- ✓ Attach your **report t**o the MS Team assignment and **turn it in**
- ✓ Present your research outcomes to the class

Website to should start your researches from!

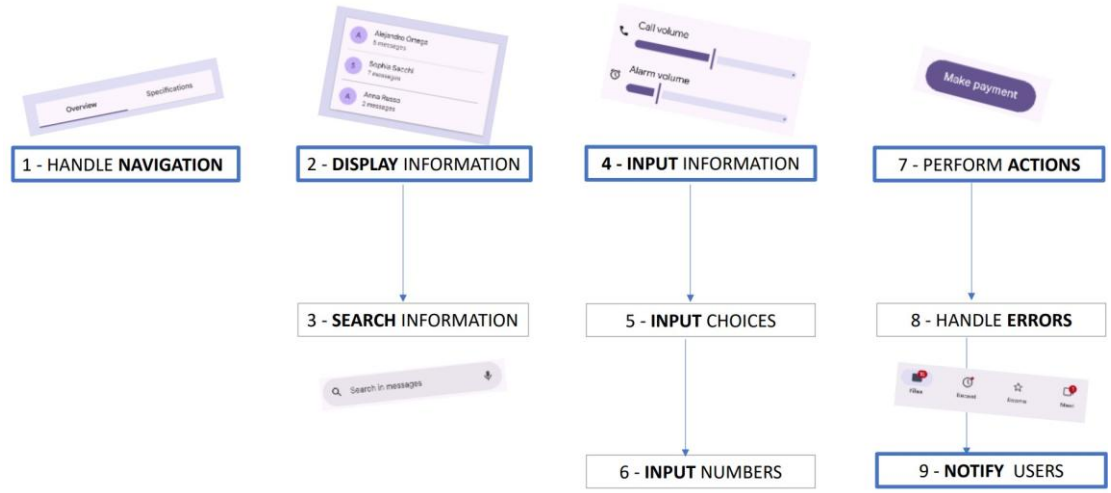
Material Design Patterns
https://m1.material.io/patterns/navigation.html

User Interface Design pattern https://ui-patterns.com/patterns

Design-Pattern Guidelines
https://www.nngroup.com/articles/design-pattern-guidelines

A - READ UI HEURISTICS

The first step of this activity is to take time to **read UI heuristics resources**:



1 - HANDLE **NAVIGATION**

COMPONENTS
https://m3.material.io/components/bottom-app-bar
https://m3.material.io/components/navigation-bar
https://m3.material.io/components/navigation-drawer
https://m3.material.io/components/navigation-rail
https://m3.material.io/components/tabs/overview
https://m3.material.io/components/dialogs/overview
https://m3.material.io/components/top-app-bar/overview

HEURISTICS https://m1.material.io/patterns/navigation.html
https://m1.material.io/patterns/navigation.html#navigation-combined-patterns
https://www.smashingmagazine.com/2017/05/basic-patterns-mobile-navigation https://ui-patterns.com/patterns/modal-windows
https://www.storyly.io/post/basic-patterns-for-mobile-navigation-and-the-best-practices

2 - **DISPLAY** INFORMATION

COMPONENTS
https://m1.material.io/patterns/empty-states.html
https://www.nngroup.com/videos/accordions-on-mobile/?lm=mobile-input-fields&pt=youtubevideo
https://ui-patterns.com/patterns/cards https://www.nngroup.com/articles/infinite-scrolling-tips/
https://ui-patterns.com/patterns/ContinuousScrolling https://ui-patterns.com/patterns/Pagination
https://ui-patterns.com/patterns/Carousel
https://www.nngroup.com/videos/avoid-accordions/

HEURISTICS
https://m3.material.io/components/lists/overview
https://m3.material.io/components/cards/overview
https://m3.material.io/components/bottom-sheets/overview
https://m3.material.io/components/carousel/overview

4 - **INPUT** INFORMATION

COMPONENTS
https://m1.material.io/components/steppers.html
https://m3.material.io/components/bottom-sheets/overview
https://m3.material.io/components/carousel/overview

HEURISTICS https://www.nngroup.com/articles/wizards/
https://blog.mobiscroll.com/building-great-mobile-forms/
https://www.nngroup.com/videos/mobile-input-fields/
https://ui-patterns.com/patterns/expandable-input
https://ui-patterns.com/patterns/WYSIWYG

5 - INPUT **CHOICES**

COMPONENTS
https://m3.material.io/components/checkbox/overview
https://m3.material.io/components/switch/overview
https://m3.material.io/components/radio-button/overview
https://m3.material.io/components/chips/overview

HEURISTICS https://www.nngroup.com/articles/toggle-switch-guidelines/
https://www.nngroup.com/articles/checkboxes-vs-radio-buttons/

6 - INPUT **NUMBERS**

COMPONENTS
https://m3.material.io/components/text-fields/overview
https://m3.material.io/components/sliders/overview
https://m3.material.io/components/chips/overview

HEURISTICS https://www.nngroup.com/articles/input-steppers/ https://www.nngroup.com/articles/sliders-knobs/
https://www.nngroup.com/articles/mobile-input-checklist/
https://www.nngroup.com/videos/mobile-input-fields/

7 - PERFORM **ACTIONS**

COMPONENTS
https://m3.material.io/components/buttons/overview https://m3.material.io/components/floating-action-button/overview https://m3.material.io/components/icon-buttons/overview https://m3.material.io/components/segmented-buttons/overview

HEURISTICS https://uxdworld.com/design-accessible-buttons/

8 – HANDLE **ERRORS**

COMPONENTS
https://m3.material.io/components/buttons/overview https://m3.material.io/components/floating-action-button/overview https://m3.material.io/components/icon-buttons/overview https://m3.material.io/components/segmented-buttons/overview

HEURISTICS https://m1.material.io/patterns/errors.html
https://www.nngroup.com/articles/hostile-error-messages/
https://www.nngroup.com/videos/mobile-error-handling-alerts/
https://www.nngroup.com/articles/confirmation-dialog/

9 - NOTIFY **USERS**

COMPONENTS
https://m1.material.io/patterns/notifications.html
https://m3.material.io/components/badges/overview
https://m3.material.io/components/dialogs/overview
https://m3.material.io/components/snackbar/overview

HEURISTICS https://m1.material.io/patterns/notifications.html https://uxplanet.org/how-to-craft-mobile-notifications-that-users-actually-want-7b585e0e1fa1
https://www.toptal.com/designers/ux/notification-design

UI HEURISTICS – WRAP UP

USE CASE	RELATED UI COMPONENTS	QUESTIONS TO CONSIDER
1- Handle Navigation	Navigation Bar App Bar Bottom bar Drawer / Navigation Rail Tabs Contextual Menu Dialog	*How important is the task?* *How frequent is the task?* *Does the task should be put with other tasks?* *Should we interrupt current user's task?* *Should we catch the user's full attention?* *How deep in the navigation tree?*

2-Display Information	Cards Tables and Lists Bottom Sheet Accordion Carousel	*How much data to display?* *How many items per screen?* *Consider the empty State pattern* *Consider infinite scroll or pagination*
3-Search Information	Free text to search Components to filter Table to sort Grouping items	*Tolerance to typos, synonyms?* *Multiple filters?* *Can user remove the filter settings?*
4-Create Information	Forms Wizards/ Steppers Bottom Sheet Accordion	*Does the data to enter must follow a process?* *Can we identify optional information?* *Should default value be provided*
5-Input choices	Checkbox Switch Radio Button Dropdown Chips	*How many choices are available?* *How many selections can the user make?* *Is there a default option?* *Are the choices depending on each other?*
6- Input numbers	TextField Slider Dropdown Chips	*How many numbers are available?* *Can many numbers be selected?* *Is there a default option?*
7- Perform Actions	Call to Action Button Secondary Button Floating Button	*What is most important action? Secondary actions?* *Is the action recoverable?* *Is the action always available?*

8-Handle Errors	Error Labels Snack bar Dialogs	*When validation should error?* *(On edition, on submit, after submitted.)* *What is the level of error?*
9- Inform Users	Badges Dialogs Push Notifications (external) In-App Banners Snack bar	*Does the information require to interrupt the user?* *Does the information need to stay?*

B - APPLY UI HEURISTICS

In this practice you will have to work on 2 scenarios (see below)

For **each scenario** you need to:

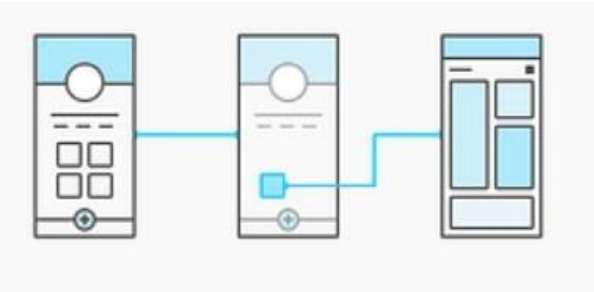
1- IDENTITY UI COMPONENTS

Identify **UI components** you want to choose, which **criteria** you **consider** and the **heuristic**s that justify your choice (see the heuristics on PART-A).

UI COMPONENTS	CRITERIA	HEURISTIC
Checkbox	Many agreements can be selected	Use checkboxes (instead of switches or radio buttons) if multiple options can be selected from a list
Radio Button	Only one option can be selected (ID type, attendance day)	Use radio buttons when choices are mutually exclusive
Text Field	User must enter free-text or numeric information (name, ID, tax number)	Match between system and real world by allowing natural data entry
Dropdown	User selects from a predefined list (country, sessions, insurance)	Error prevention by restricting input to valid options
Segmented Control	User switches between exclusive modes (Full / Multiple Day / Online)	Consistency and standards for switching related views
Date Picker	User selects a valid date (date of birth, attendance date)	Error prevention by avoiding incorrect date formats
Button (Primary)	User performs main action (Next, Confirm payment)	Recognition rather than recall through clear call-to-action
Icon Button	User navigates to previous screen	User control and freedom to undo or go back
Card	Group related information (payment summary, review details)	Aesthetic and minimalist design by organizing content

Dialog	User must confirm irreversible actions (payment)	Error prevention by requesting confirmation
Inline Error Message	Input is invalid or missing	Help users recognize, diagnose, and recover from errors

2- DESIGN THE WIREFLOW



Provide:

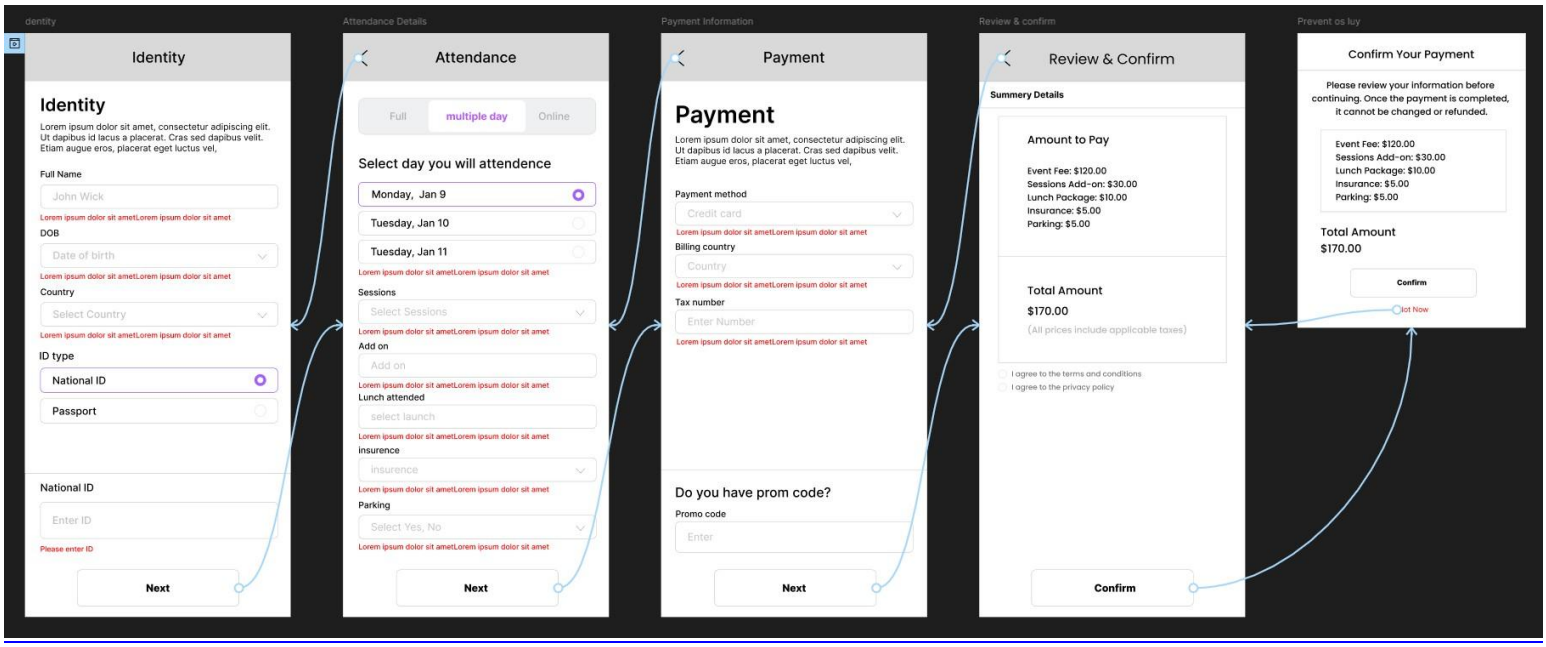
- Screen-to-screen wireflow
- Navigation transitions
- Error states
- Alternative paths
- Back navigation behavior

C – SCENARIO PRESENTATION

International Event Registration & Access Pass

As a User, register for a **multi-day international event** (conference / festival / competition).

Identity	
Full name	
Date of birth	(18 years or greater)
Nationality	
ID type (Passport / National ID)	Some countries have different ID formats
Attendance Details	
Attendance type	Full / multiple day / online
Days attending	In case of multiple days - event can have multiple days
Sessions	80+ choices - only the ones that applies
Add-Ons	
Lunch attended	Shall be compliant with the attendance
Insurance	Insurance required for foreign attendees
Parking	
Payment Information	
Payment method	

Billing country	
Tax number	optional / required depending on country
Promo code	
Review & confirm	
Summary	
Legal agreements	Many agreements shall be approved
Payment confirmation	

- WireFrame